ÇUKUROVA UNIVERSITY

ENGINEERING FACULTY

DEPARTMENT OF COMPUTER ENGINEERING

Operating Systems

Second Assignment

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1. Scheduling Algorithms Implementation

Abstract

In this report, I will write about scheduling algorithms and their C language implementation.

a. First-come, first-served Scheduling(FCFS)

FIFO sorts the process in queue order. The first action is executed, and then the next action starts after the previous action has been completely executed. The arrival time of all processes is assumed to be 0. [1]

b. Shortest-job first Scheduling(SJF)

It has the advantage of having the shortest total waiting time. It is practically impossible to use as the operating system does not know the burst time and therefore cannot enumerate them. SJF can only be used in special cases where uptime estimates exist. [2]

Implementation:

- Sort all processes by their arrival time.
- Choose the method with the shortest arrival and explosion time.
- Create a process pool that will run until the previous process is complete and select the process with the shortest Burst time from the pool. [2]

c. Priority Scheduling(P)

It is very popular. Every operation has a priority. The highest priority process is carried out first. Prioritization can be determined based on memory requirements, time constraints or other resource constraints. [3]

Implementation [3]:

- Enter the time and priority of the detonation of processes.
- Organize procedures according to explosion time and priority.
- Use the FCFS algorithm.

d. Round-robin Scheduling(RR)

Since all processes have an equal share in the CPU, it takes a short time to set up a programming algorithm that does not cause CPU hunger. Basically it is a widely used method in CPU scheduling. Preventive because processes only throw CPU for a limited time. [4]

Implementation[4]:

- Completion Time: Time at which process completes its execution.
- Turn Around Time: Difference in time between completion-time and arrival-time.
- Turn Around Time = Completion Time – Arrival Time
- Waiting Time (W.T): Difference in time between turn-around and burst times.
- Waiting Time = Turn Around Time – Burst Time

e. Priority with Round-robin Scheduling(PRR)

Each process has a priority, but if the process takes longer than the selected CPU time, the algorithm moves to execute the next program with the highest priority.

2. Implementation Details

I used CodeBlocks IDE in Ubuntu. From Terminal I typed “CodeBlocks” and application started. I created a project and added all library files (.h extension). Then I wrote the wanted functions for each scheduling algorithm. You have to add input files (.txt extension) that represent the processes. Later on build each scheduling algorithm independently from other scheduling algorithms because otherwise their function names are overlapping and giving error.

Youtube Link: https://youtu.be/Zh47NkBLYqw

Conclusion

Many different scheduling algorithms exit and all have their own pros and cons. Some of them are theorical and not useful in practice because we don’t know command order or priority in real situations. These algorithms provide convenience during the procedures.

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